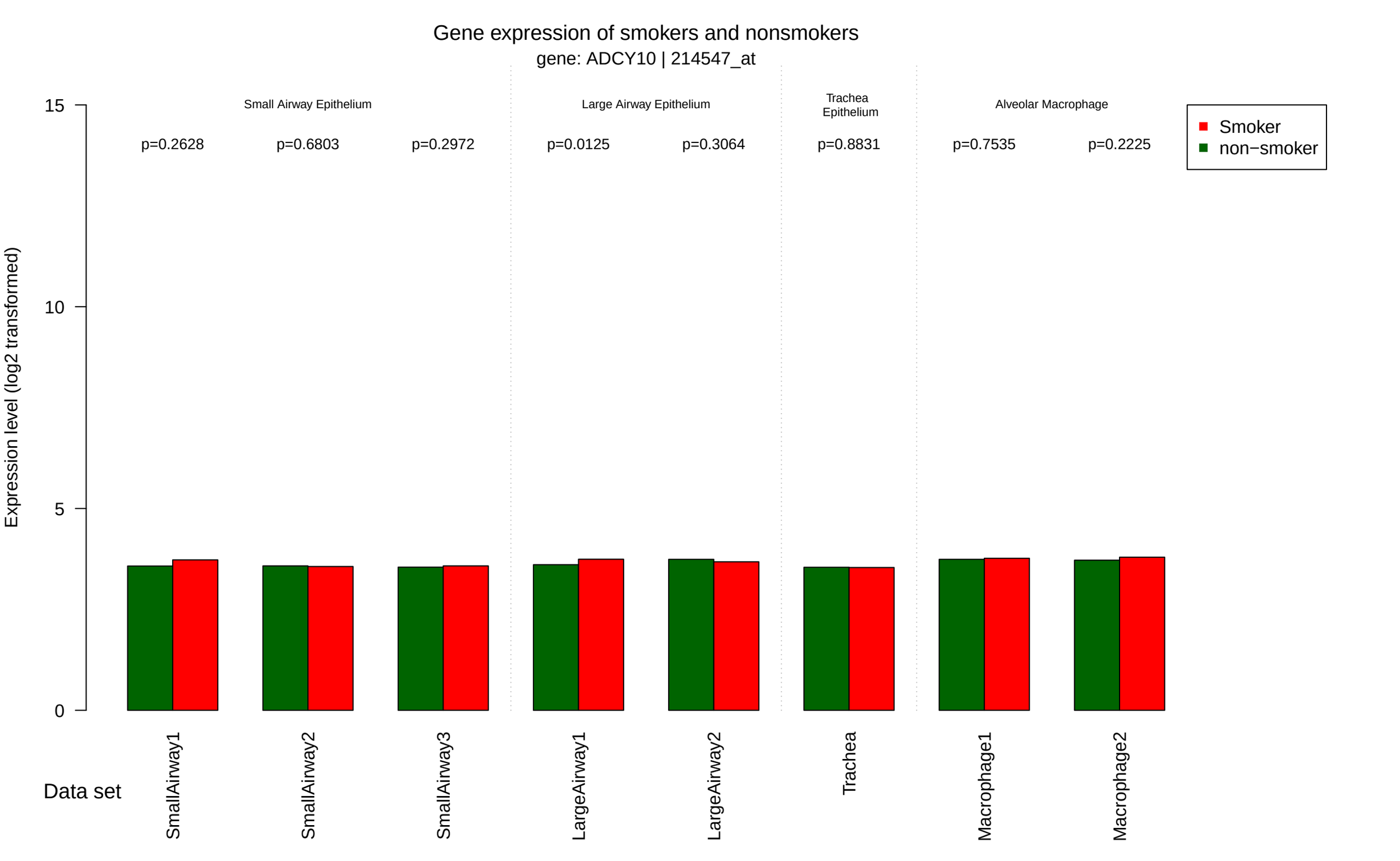
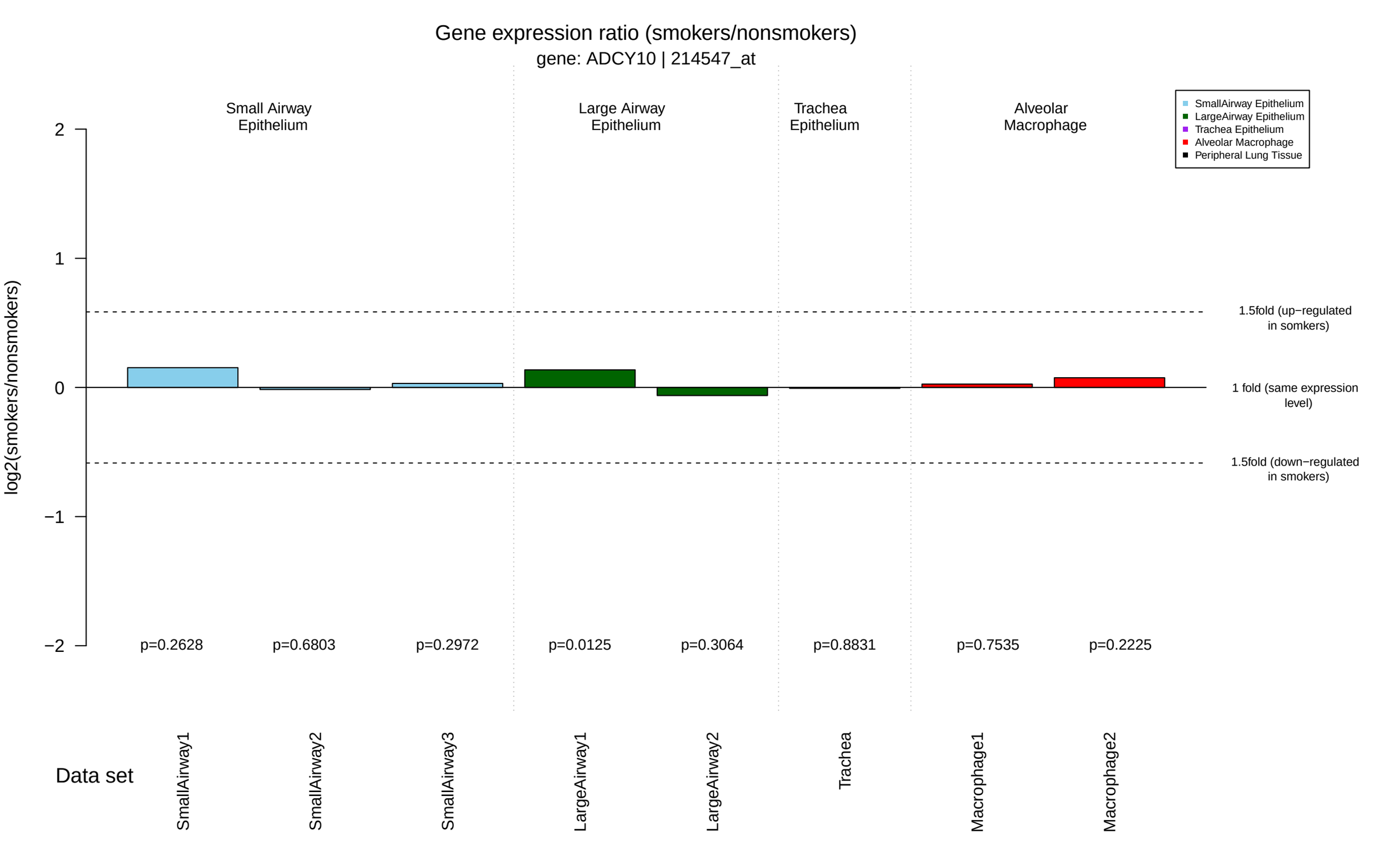
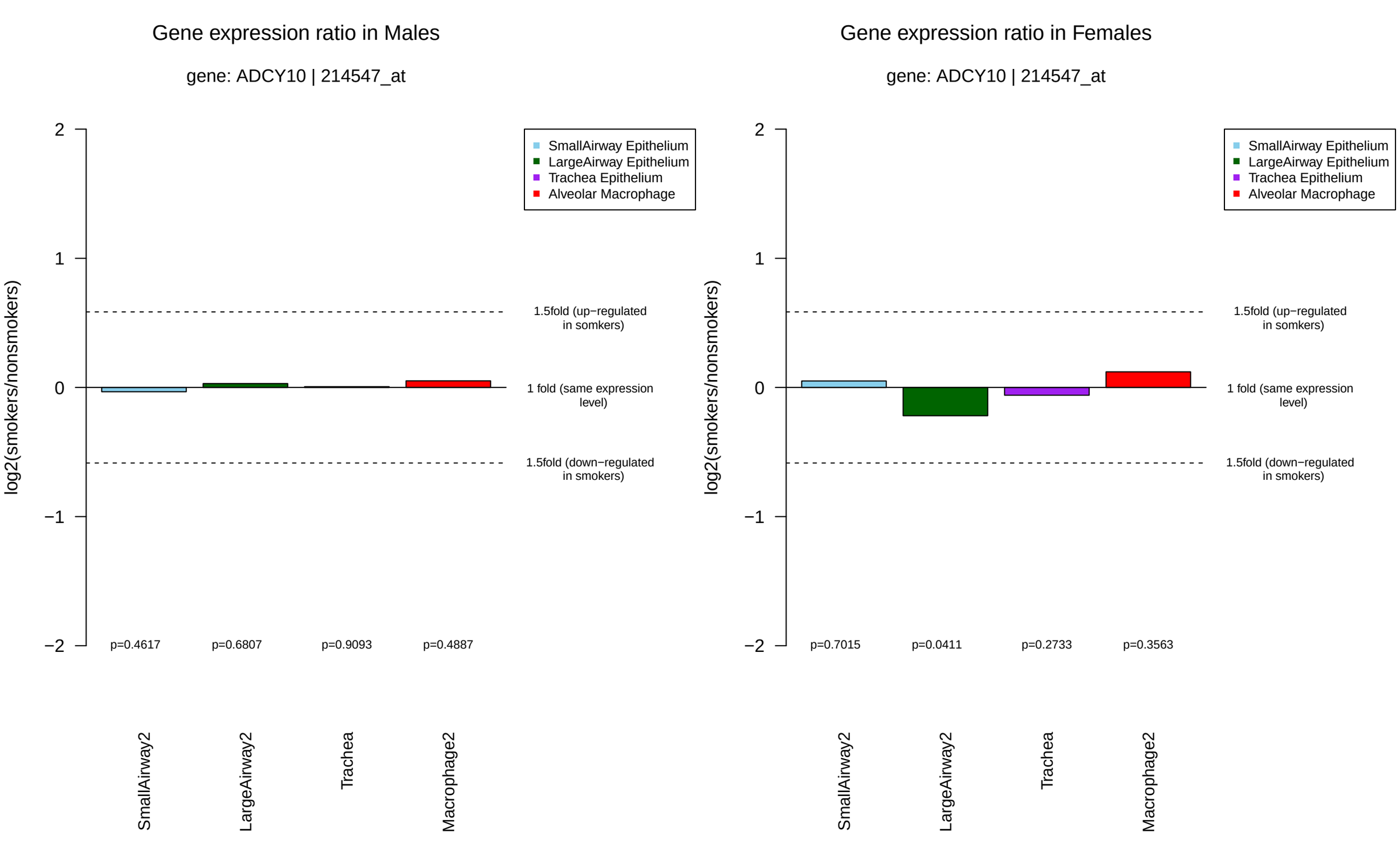
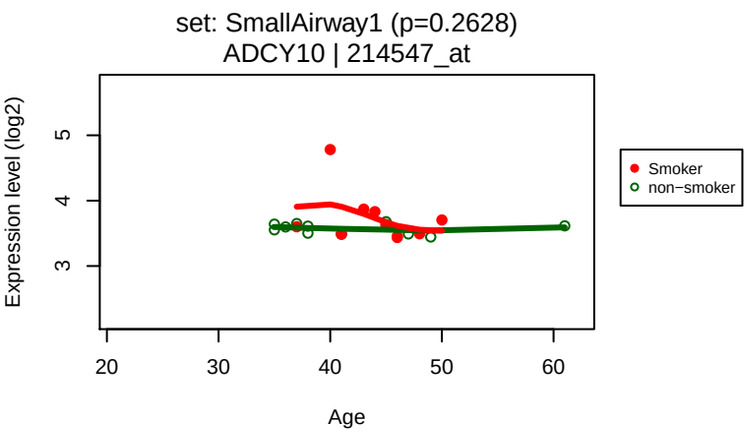
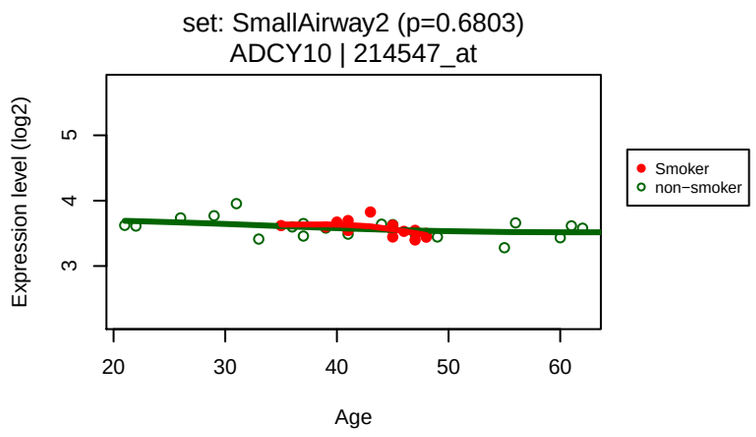
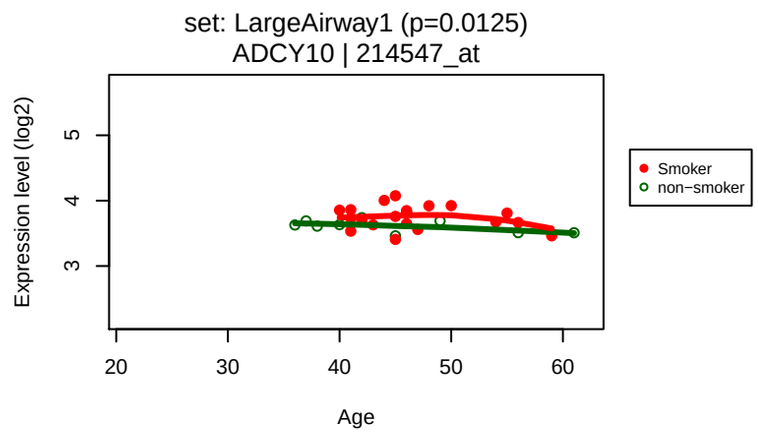

Gene expression of smokers and nonsmokers
gene: ADCY10 | 214547_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2628
p=0.6803
p=0.2972
p=0.0125
p=0.3064
p=0.8831
p=0.7535
p=0.2225
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

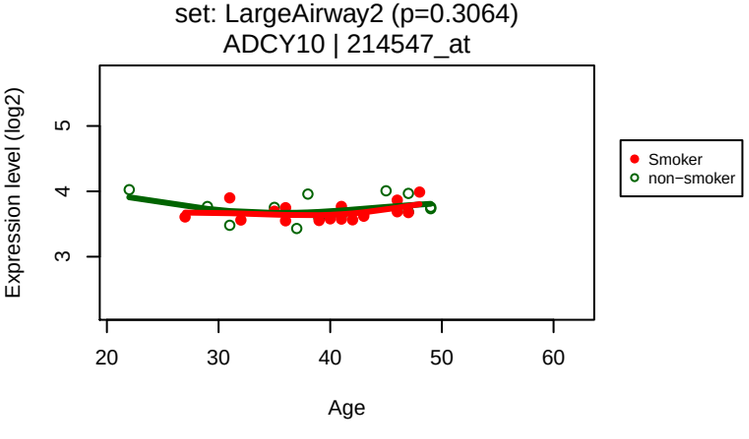

Gene expression ratio (smokers/nonsmokers)
gene: ADCY10 | 214547_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2628
p=0.6803
p=0.2972
p=0.0125
p=0.3064
p=0.8831
p=0.7535
p=0.2225
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

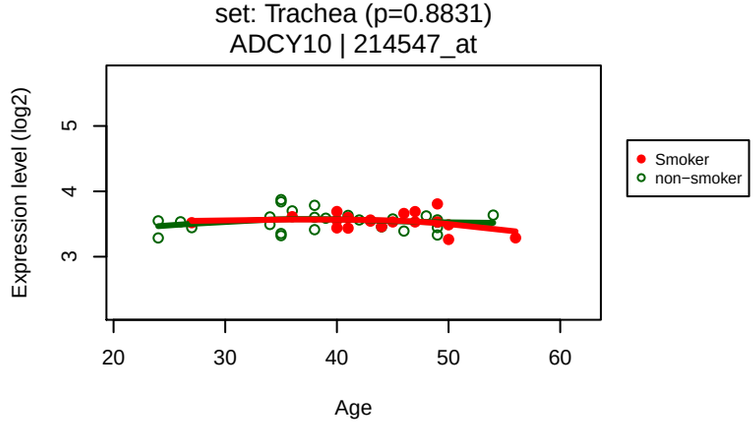

Gene expression ratio in Males
gene: ADCY10 | 214547_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4617
p=0.6807
p=0.9093
p=0.4887
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ADCY10 | 214547_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7015
p=0.0411
p=0.2733
p=0.3563
SmallAirway2
LargeAirway2
Trachea
Macrophage2

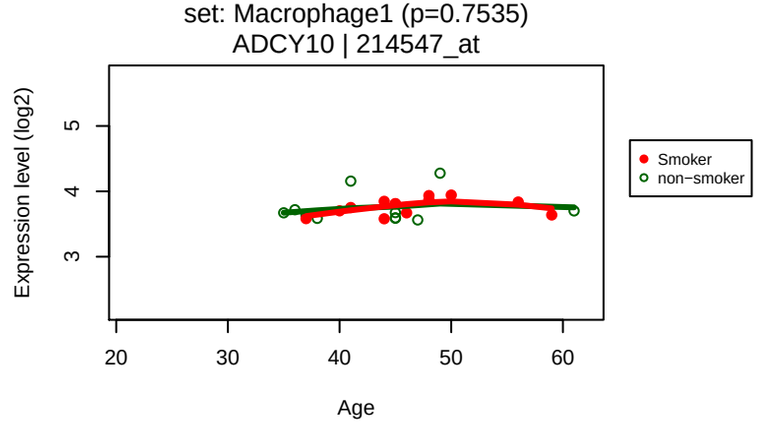

set: SmallAirway1 (p=0.2628)
ADCY10 | 214547_at
Expression level (log2)
Age
Smoker
non−smoker

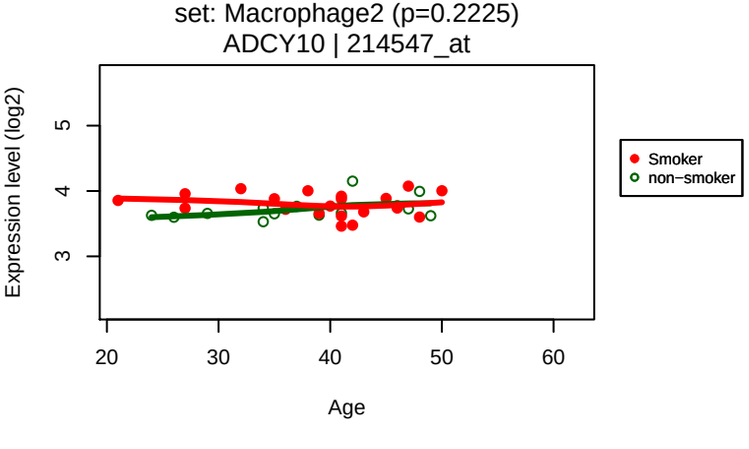

set: SmallAirway2 (p=0.6803)
ADCY10 | 214547_at
Expression level (log2)
Age
Smoker
non−smoker

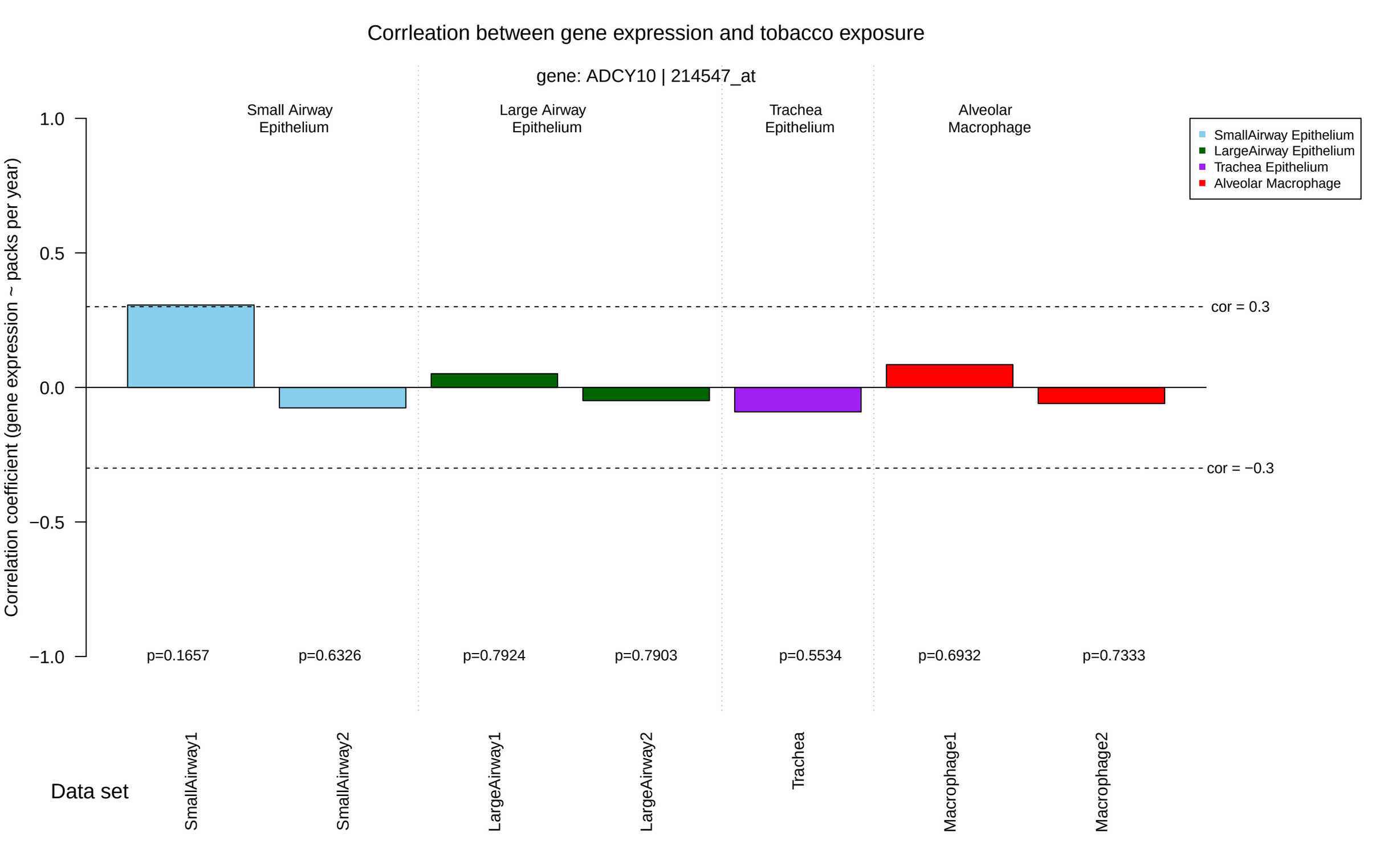

set: LargeAirway1 (p=0.0125)
ADCY10 | 214547_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3064)
ADCY10 | 214547_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8831)
ADCY10 | 214547_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7535)
ADCY10 | 214547_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2225)
ADCY10 | 214547_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ADCY10 | 214547_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1657
p=0.6326
p=0.7924
p=0.7903
p=0.5534
p=0.6932
p=0.7333
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2